

Client Connection Guide

Connect stock Element or Element X and complete the first encrypted messaging test

v0.3.0 Offline Release | Windows 11 | July 2026

Item	Value
Homeserver URL	Environment-specific (POC: http://10.20.30.2)
Matrix identity	axon.home.arpa
Client networks	Flat LAN or routed/NATed private subnets
Host exposure	TCP 80 only

Axon provides text messaging through stock Element or Element X. Internet, cellular data, federation, media, calling, and internet push notifications are outside the initial proof of concept.

1. Reach Axon

- Install Element or Element X from an approved store or sideload source.
- Join the intended router/client network and obtain a unique address from that router's DHCP server.
- Disable AP/client isolation and confirm the environment-specific Axon address is reachable through the intended route, NAT, or forward.
- No internet, cellular data, public DNS, public certificate, or push service is required for the initial test.

```
# Replace with the Axon address visible to this client
Test-NetConnection 10.20.30.2 -Port 80
curl.exe --no-proxy "*" http://10.20.30.2/_matrix/client/versions
```

2. Sign in

- Choose Sign in, then Change or Edit homeserver.
- Enter the operator-issued URL exactly, for example `http://10.20.30.2`.
- Enter the administrator-created username and password. The full identity is `@username:axon.home.arpa`.
- Accept any client warning that internet features or push notifications are unavailable; do not substitute `matrix.org`.
- After first login, change any issued stock password in account/security settings. Axon requires at least 10 characters for GUI-issued passwords.

HTTP compatibility varies by client build and mobile operating-system policy. Test the current stock applications first. If a build refuses cleartext HTTP, record the exact app version and error; the planned fallback is private-CA TLS, not public internet TLS.

3. Two-client messaging test

- Sign in as separate users on Client A and Client B.
- Create a private encrypted room, invite the other Axon identity, and accept the invitation.
- Send A to B, B to A, then disconnect B and send another message from A.
- Reconnect B within 48 hours and confirm synchronization.
- Verify unrelated application traffic independently. Axon only hosts its documented Matrix services.

Notification expectation

Synapse push processing is disabled. Messages synchronize when the client is active or reconnects. Axon purges eligible server-side history after the configured lifetime; E2EE hides plaintext but does not eliminate temporary encrypted event storage.

Troubleshooting order

Symptom	Check
No TCP connection	Client subnet, isolation, Windows NIC, firewall, and link state
TCP succeeds, URL fails	Run Matrix versions URL; confirm all three containers are healthy
URL accepted, login spins	Synapse logs, exact credentials, and client HTTP policy
No background alert	Expected without internet push; reopen app and resynchronize